

Mechanical Engineering Design Portfolio

Bereket Zemedkun

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Welcome, and thank you for taking the time to view my portfolio. The goal of this portfolio is to give you a deeper insight into my experiences and skills I have gained over my recent history.



It is my hope that this will allow you to better assess how my skills can be applied to your company.



I would be happy to talk in more detail and can be reached using the contact information on the previous page.

Certifications

CSWP: https://www.credly.com/badges/be7599ce-0f94-4338-8c7a-479d6bd44243/public_url



CSWA:https://www.credly.com/badges/6cc2df2e-bb73-4624-b450-319663b05c2e/public_url



CERTIFICATE

Dassault Systèmes confers upon
BEREKET ZEMEDKUN
the certificate for
SOLIDWORKS CAD Design Associate



November 26 2025



C-48UPQ9TR5J

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

CSWP-DT: https://www.credly.com/badges/de133b49-3537-4c21-a883-dc31c6c4ae80/public_url



CERTIFICATE

Dassault Systèmes confers upon
BEREKET ZEMEDKUN
the certificate for
SOLIDWORKS Drawing Tools Professional



December 22 2025



C-N2QLJ3K8X7

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

SOLIDWORKS Electrical Design Associate : https://www.credly.com/badges/b3d131f8-90f1-4ddd-a7e5-267b14fe6ff6/public_url



CERTIFICATE

Dassault Systèmes confers upon
BEREKET ZEMEDKUN
the certificate for
SOLIDWORKS Electrical Design Associate



December 31 2025



C-TPYRW8L9UD

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

SOLIDWORKS Additive Manufacturing Associate :

https://www.credly.com/badges/474be813-81dd-4769-be45-e65fac65158b/public_url



CERTIFICATE

Dassault Systèmes confers upon
BEREKET ZEMEDKUN
the certificate for
SOLIDWORKS Additive Manufacturing Associate



December 12 2025



C-B4NK7Q3SYE

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

SOLIDWORKS Sustainability Associate :



CERTIFICATE

Dassault Systèmes confers upon
BEREKET ZEMEDKUN
the certificate for
SOLIDWORKS Sustainability Associate



December 27 2025

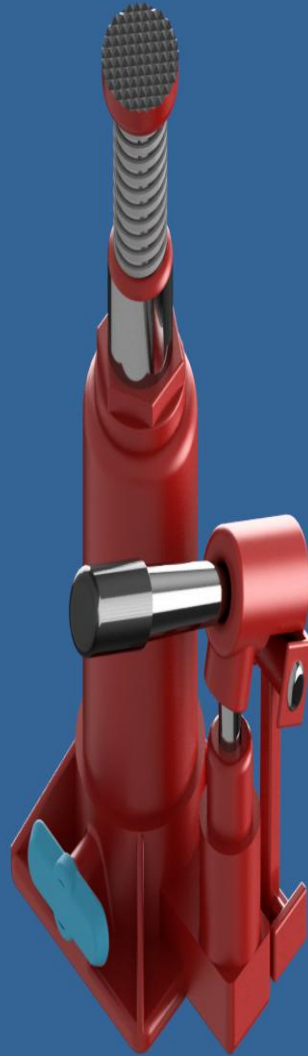


C-7JH6QLKGSY

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

HYDRAULIC BOTTLE JACK Project



- Role: Mechanical Design and Design Student
- Date: Sep 2024 – Feb 2025
- Skills: Mechanical Design, CAD (Solidworks)

HYDRAULIC BOTTLE JACK

- Aim: To design and develop a hydraulic bottle jack that can efficiently lift heavy loads with minimum manual effort by utilizing hydraulic pressure principles, ensuring safety, strength, stability, and durability of all components.

Requirements:

1. **Load Capacity** – Lift intended heavy loads safely.
2. **Hydraulic System** – Efficient cylinder, piston, and valves for smooth operation.
3. **Material Strength** – Durable, corrosion-resistant, high-strength components.
4. **Stability & Safety** – Wide base and secure design to prevent tipping.

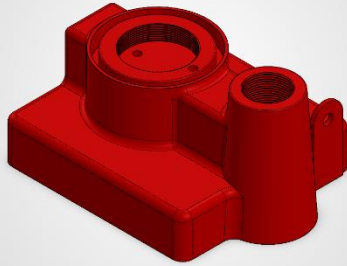
Result:

- The hydraulic bottle jack successfully lifted the intended load.
- **Analysis was performed using ANSYS** to simulate stress, deformation, and safety under maximum load.
- Results showed all components withstand hydraulic pressure without failure.
- Design meets stability, strength, and performance requirements.

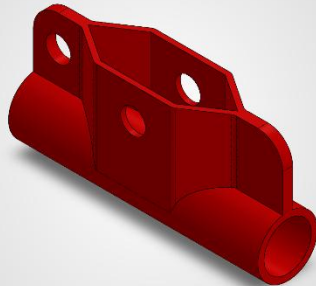




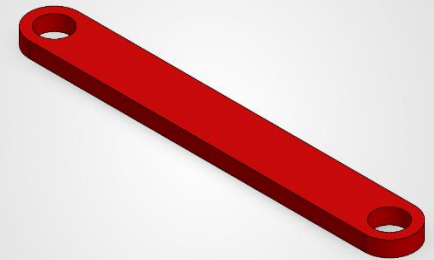
Part Files



Base



Pump piston set



Pump lever

Porsche 911 GT3 Project (Personal project)

Role: Mechanical Design and Surface
modeling

Date: April 2025 – June 2025

Skills: Mechanical Design, CAD (SolidWorks
,keyshot(rendering)



Porsche 911 GT3 Project

- Aim: The aim of this project is to create a precise and realistic **3D model of the Porsche 911 GT3** using advanced surface modeling techniques in SolidWorks. The project focuses on capturing the complex curves and surfaces of the car's body accurately, ensuring the model is high-quality, parametric, and suitable for rendering, visualization, or further design applications.

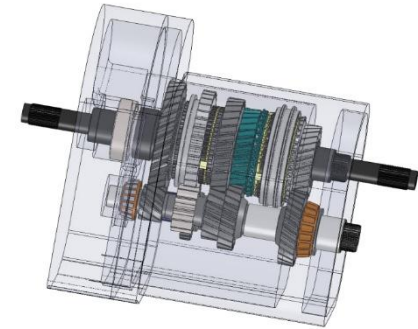
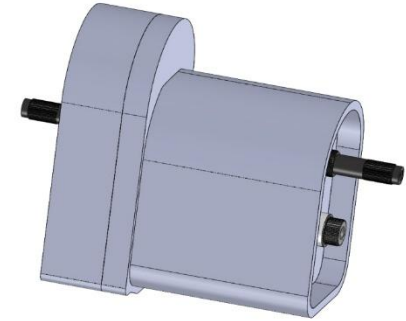
Result:

- Successfully created the **Porsche 911 GT3 3D model** with precise, smooth, and continuous surfaces.
- Produced **high-quality realistic renderings in KeyShot** showcasing the car's design accurately.



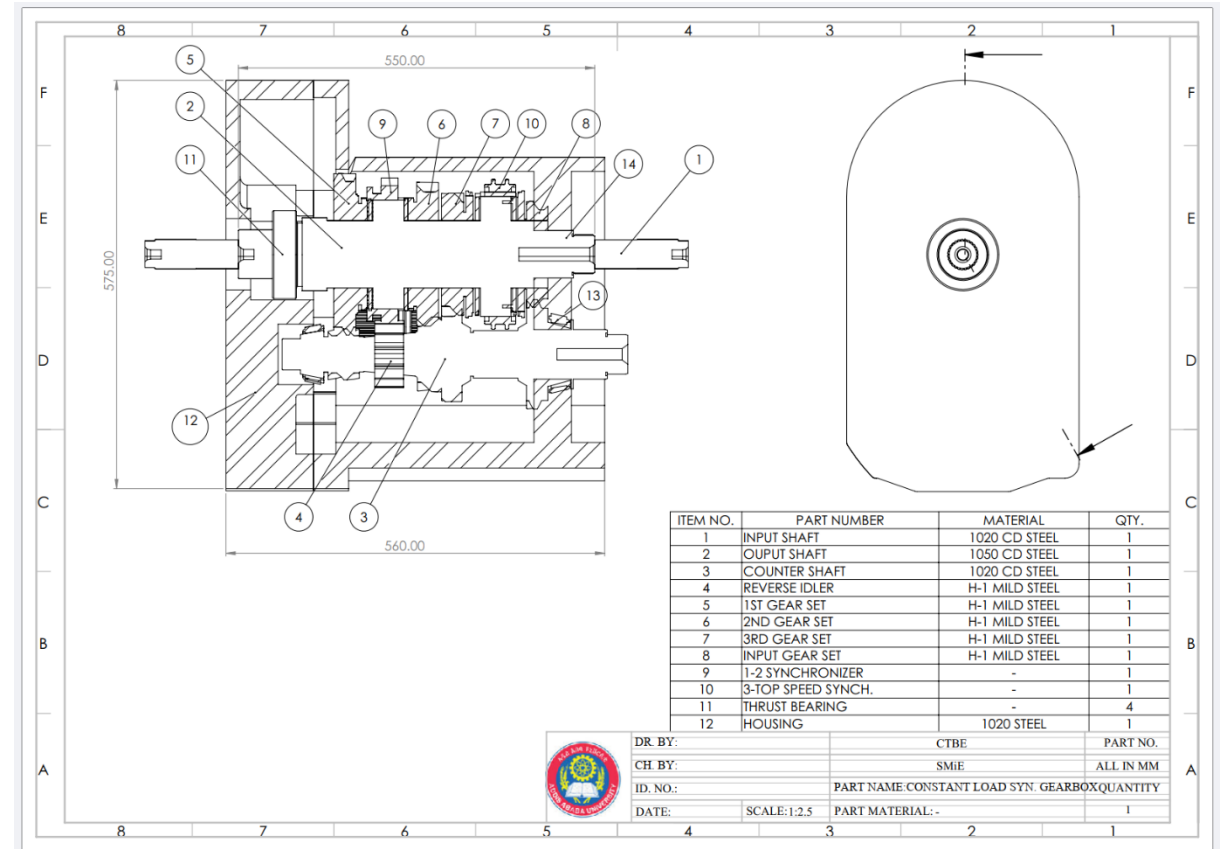
Gear Box Design

- Role: Mechanical Design and Manufacturing Student
- Date: Sep 2025 – Jan 2026
- Skills: Mechanical Design, CAD (Solidworks)



Gear Box Design

- Aim: Design the three-shaft gear-train system in a housing separated at the plane of shaft centerlines. The transmission is expected to work for at least 8 years .
- Result: Successfully designed one shaft of a complete three-stage gear box including the assembly and part drawings and analyzed the safety of the overall system



HAMMER MILL Re-Design

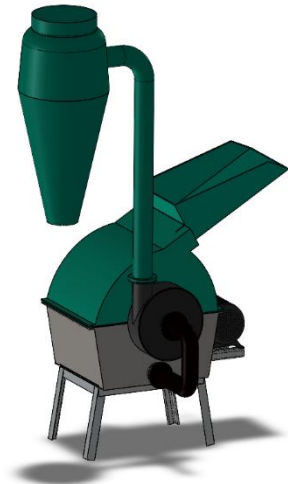


Role: Mechanical design and
mechanical engineer

Date: Sep 2025 –Dec 2025

Skills: Mechanical Design,
CAD (SOLIDWORKS),
Research, FEA

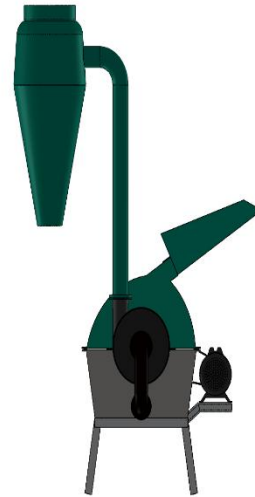
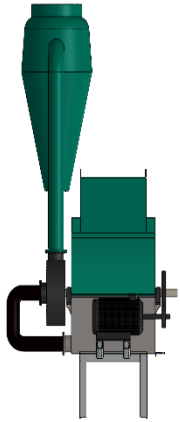
Project Background



- Aim: The aim of this project is to reverse engineer an existing hammer mill and redesign it by adding additional hammers to improve grinding efficiency, enhance performance, and develop an accurate CAD model for design improvement and manufacturing.
- Result: The reverse engineering and redesign of the hammer mill were successfully completed, and an accurate CAD model of the existing machine was developed. Additional hammers were incorporated into the rotor assembly while maintaining proper spacing and balance.

As a result, the modified hammer mill design showed the following improvements:

- Increased grinding efficiency due to more impact points
- Reduced grinding time for the same quantity of material
- Improved uniformity of particle size
- Better utilization of rotor rotation and power
- A complete and detailed CAD model suitable for manufacturing and further analysis



Electrical Scooter Design Project

AIM: The main aim of this school project is to design and develop the mechanical structure of an electrical scooter using CAD software and to evaluate its structural strength using Finite Element Analysis (FEA). The project focuses on designing key mechanical components such as the frame, motor mount, wheel support, and battery holder to ensure the scooter is strong, stable, and safe.

In the context of Ethiopia, this project also aims to develop mechanical design solutions that can be applied to affordable and efficient transportation systems suitable for local conditions. The project helps in understanding load distribution, structural behavior, and proper mechanical assembly of electric vehicles.



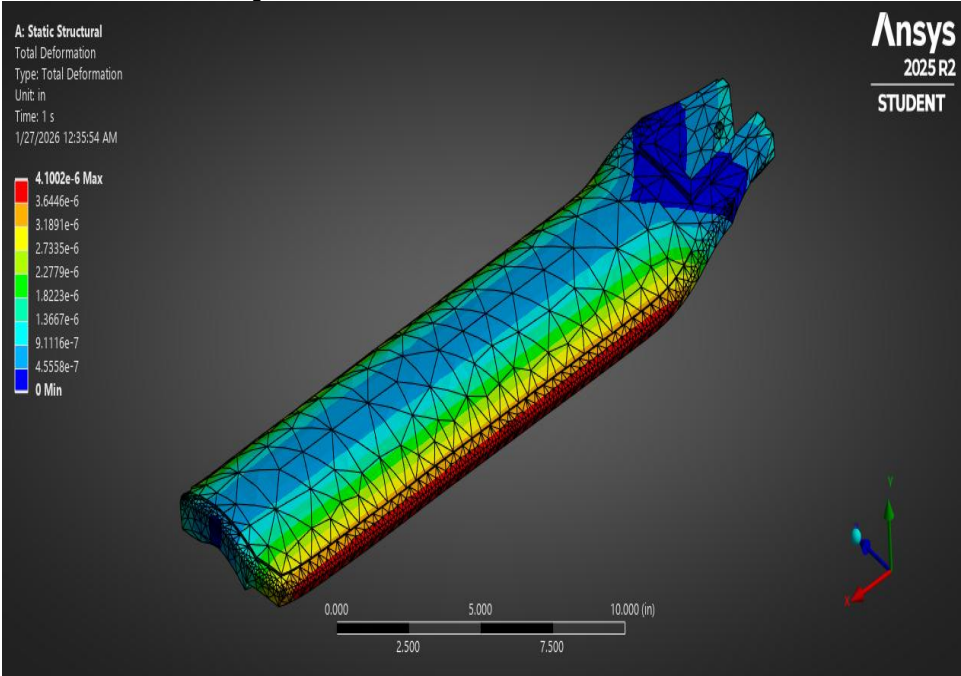
Result of Electrical Scooter Design Project (Mechanical Design with FEA)

As a result of this project, a complete mechanical design and 3D CAD assembly of the electrical scooter was successfully created. The frame, motor mount, and wheel support components were properly designed and assembled to ensure structural stability and proper alignment.

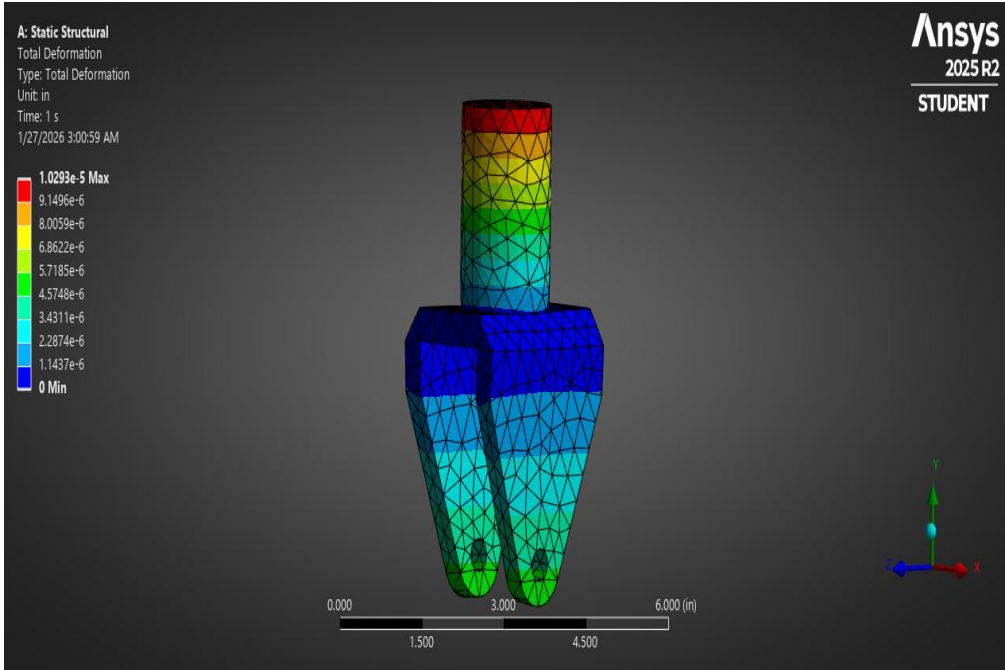
Finite Element Analysis (FEA) was performed on the scooter frame to evaluate its strength under loading conditions. The analysis results showed that the stress and deformation were within safe limits, confirming that the frame design is structurally safe and reliable.

Structural analysis by using ANSYS

Stress analysis for Deck



Stress analysis for Front Fender:



Advanced Surface modeling of F1helmet

Role: Personal Project

Date: Nov 2025

Skills: Mechanical Design,
Surface modeling, CAD
(SOLIDWORKS), Keyshot
(Rendering)



F1 helmet Project

Aim: The aim of this personal project is to design a Formula 1 (F1) helmet using advanced surface modeling techniques in CAD software. The focus is on developing a smooth, aerodynamic, and precise surface geometry that reflects real-world F1 helmet shapes. This project enables practical learning of advanced surface tools, complex curve management, and precision modeling, which are critical skills for high-end product and automotive design. The project aims to enhance proficiency in CAD surface modeling by creating a visually accurate and manufacturable 3D helmet model, helping to bridge the gap between theoretical knowledge and practical advanced modeling skills.

Result: As a result of this personal project, a complete 3D CAD model of the F1 helmet was successfully created using advanced surface modeling techniques. The helmet features smooth, precise, and aerodynamic surfaces, demonstrating mastery of complex surface tools and curve manipulation in CAD software.

The final model was also rendered in **KeyShot**, producing realistic visuals with accurate materials, reflections, and lighting. This helped visualize the helmet as it would appear in a real-world scenario, enhancing the presentation quality of the project.

Overall, the project improved skills in advanced surface modeling, CAD design precision, and professional rendering, providing a strong foundation for future high-end product design and visualization projects.



Fig: The final F1 helmet project